BEU CONNECT

Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025

Time: 03 Hours

Code: 100211 ***Subject: Workshop*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

(a) GI sheet refers to:
- (i) Galvanized iron
- (ii) General-purpose iron
- (iii) Glass-insulated sheet
- (iv) Grey iron

(b) Micrometer screw gauge is used for:
- (i) Large distances
- (ii) Cutting sheet
- (iii) Internal diameters
- (iv) Small and precise thickness

(c) Which of the following is a solid-state joining process?
- (i) Brazing
- (ii) TIG welding
- (iii) Friction welding
- (iv) Soldering

(d) Which gas combination is most common in gas welding?
- (i) Hydrogen-Oxygen
- (ii) Propane-Air
- (iii) Acetylene-Oxygen
- (iv) Nitrogen-Oxygen

(e) Which of the following is not a pattern material?
- (i) Wood
- (ii) Wax
- (iii) Cast iron
- (iv) Silicon carbide

(f) The purpose of a core in sand casting is to:
- (i) Strengthen mould box
- (ii) Form internal cavities in casting
- (iii) Increase permeability
- (iv) Prevent shrinkage

(g) Which file has a cross-section shaped like a triangle?
- (i) Flat file
- (ii) Half-round file
- (iii) Triangular file
- (iv) Square file

(h) A ball peen hammer is primarily used for:
- (i) Wood working
- (ii) Punching and riveting
- (iii) Welding
- (iv) Drilling

(i) The wood defect caused by unequal shrinkage is:
- (i) Shake
- (ii) Warp
- (iii) Knot
- (iv) Split

(j) The furnace used for heating metals in a smithy shop is called:
- (i) Cupola
- (ii) Hearth
- (iii) Kiln
- (iv) Reverbatory furnace

Q.2 Solve both questions : **[14]**

(a) Classify the joining processes based on state of the filler material during joining.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

(b) Explain the principle of gas welding with a neat sketch of the oxy-acetylene welding setup.

Q.3 Solve both questions : **[14]**

(a) Sketch the cross-section of a sand mould which is ready for pouring, and label the various important parts.

(b) Explain the types of pattern with schematic diagram.

Q.4 Solve both questions : **[14]**

(a) Write down the various types of holding tools, cutting tools, marking tools and measuring tools, which is used in fitting shop.

(b) Write down the types filling tools with schematic diagram.

Q.5 Solve both questions : **[14]**

(a) Explain the structure of wood with schematic diagram.

(b) What are the carpentry joints and draw them with dimension?

Q.6 Solve both questions : **[14]**

(a) What are the materials generally used in sheet metals work?

(b) Explain the sheet metal operations.

Q.7 Solve both questions : **[14]**

(a) What are the tools used in foundry shop?

(b) Explain the operation used for developing the eye nail in foundry shop.

Q.8 Solve both questions : **[14]**

(a) What are safety precaution used in workshop during operation?

(b) Describe the processes and operations taught in Workshop Technology.

Q.9 Write short notes on the following: **[14]**

(a) Sand casting

(b) Hot working process

(c) Seasoning process

(d) Arc welding

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store